

G156HAN05.0

EC Report

AUO Display Plus

2023/11/27

AUO Display Plus LED Driver IC and Power IC EC



- Purpose : Since LED driver IC (NT 50172) will be EOL. In order to keep supply smoothly, ADP will launch a replaced model, including new PCBA(LED driver IC NT50508) for longevity support.

- Change Items:

Items	Before GI56HAN05.000	After GI56HAN05.004
LED driver IC	NT50172 ,NVT	NT50508 , NVT
PCB	Old	New

AUO Display Plus LED Driver IC and Power IC EC



➤ Spec Change:

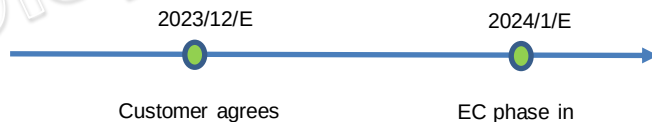
Current:

Parameter	Symbol	Min	Typ	Max	Units	Remark
LED Power Supply	VLED	10.8	12.0	13.2	[Volt]	Define as Connector Interface (Ta=25°C)
LED Enable Input High Level	VLED_EN	2.5	-	-	[Volt]	
LED Enable Input Low Level		-	-	0.8	[Volt]	
PWM Logic Input High Level	VPWM_EN	2.5	-	-	[Volt]	
PWM Logic Input Low Level		-	-	0.8	[Volt]	
PWM Input Frequency	FPWM	200	1K	10K	Hz	
PWM Duty Ratio	Duty	5	-	100	%	

New:

Parameter	Symbol	Min	Typ	Max	Units	Remark
LED Power Supply	VLED	10.8	12.0	13.2	[Volt]	Define as Connector Interface (Ta=25°C)
LED Enable Input High Level	VLED_EN	2.5	-	-	[Volt]	
LED Enable Input Low Level		-	-	0.8	[Volt]	
PWM Logic Input High Level	VPWM_EN	2.5	-	-	[Volt]	
PWM Logic Input Low Level		-	-	0.8	[Volt]	
PWM Duty Ratio (200Hz~5KHz)	Duty	5	-	100	%	
PWM Duty Ratio (5KHz~10KHz)	Duty	15	-	100	%	

➤ Plan schedule:



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Optics test



Optics Pass → OK

Optical Measurement Report

Originator	chun li	Model/PCS	G156HAN05.0/5pcs	Date	2023/5/28
Test Condition	Lum.,CR,Chromaticity,Uniformity,Gary Scale,V-T,View Angle,RT,CrossTalk				
Test System	Detector	Inverter	DC Voltage	Warm up time	
FPM-520-2	SR-UL1R	NA	12V	30 min	

【Test Reason】

G156HAN05.0 DVT1-1 RA Test

【Judgement】

PASS

Test Items	unit	PSS			23.0°C	23.0°C	23.0°C	23.0°C	23.0°C	-4σ	mean	+4σ
					2023/5/28 8:27	2023/5/28 8:52	2023/5/28 9:16	2023/5/28 9:47	2023/5/28 10:15			
					R3B35Q3XCZZS01R	R3B35Q2CCZZS01R	R3B35Q2CQZZS01R	R3B35Q2HGZZS01R	R3B35Q2FMZZS01R			
					G5100133YZ0314911	G5100133YZ0314911	G5100133YZ0314911	G5100133YZ0314911	G5100133YZ0314911			
		min	typ	max								
Center Brightness	cd/m ²	400	500		570.1	540.5	567.2	579.9	567.7	506.416	565.1	623.744
Darkness (L0)					0.41	0.38	0.41	0.40	0.39	0.347	0.40	0.451
Contrast Ratio(CR)	---	700	1000		1391.2	1415.7	1376.7	1440.0	1457.1	1283.009	1416.1	1549.268
Brightness(5P Ave.)	cd/m ²				523.8	500.6	521.3	527.6	526.3	475.660	519.9	564.164

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RA test

RA Pass → OK



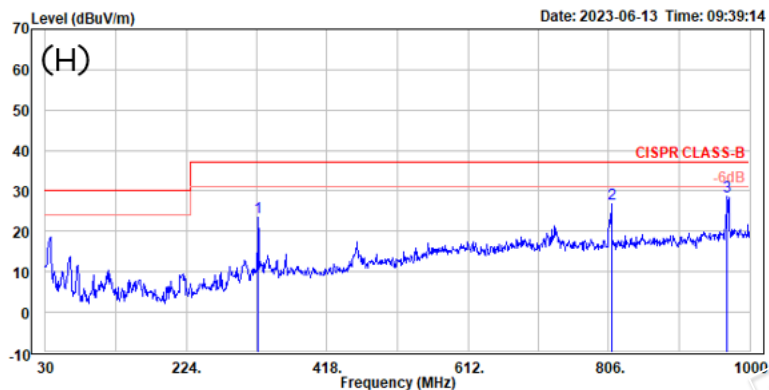
RA Test Summary Report

保管庫位 Storage	RA
保存期間 Retention Period	板面生產批號與儲存位置對照表 AU00P-03-007 系統管理維護部

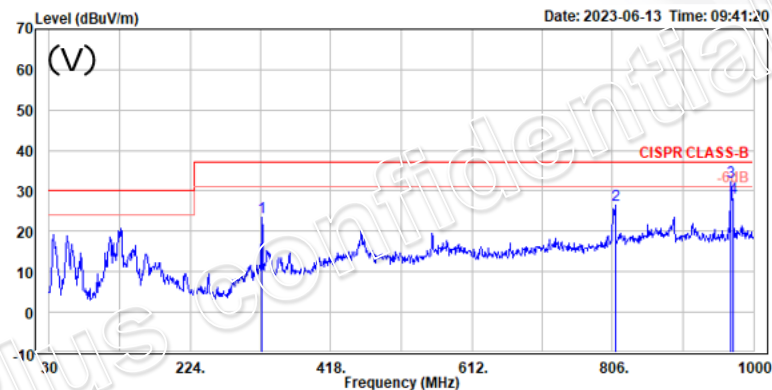
Product No	G158HAN65.0		Part No	6718G51.004		Applicant/Dept	系統部/ERP/PF3		Judgement		
Test Purpose	G158HAN65.0 DVT1-1 RA Test					Application Date	2023/05/22		PASS		
RA No	D230520008		Product Stage	DVT		Issue Date	2023/05/28				
Test Item	Test Condition	Test Phase				Test Result					
HTO	60°C 500h	No	Chip_Id	Shipping_No	T0*	T100*	T200*	T300*	T500	Judgement	Judgement
		D230520006001		6R3B36Q3UJZZZ30104	OK	OK	OK	OK	OK	PASS	PASS
		D230520006002		6R3B36Q3UBZZZ30104	OK	OK	OK	OK	OK	PASS	
		D230520006003		6R3B36Q3RRZZZ30104	OK	OK	OK	OK	OK	PASS	
LTO	-10°C 500h	No	Chip_Id	Shipping_No	T0*	T100*	T200*	T300*	T500	Judgement	Judgement
		D230520006004		6R3B36Q10FZZZ30104	OK	OK	OK	OK	OK	PASS	PASS
		D230520006005		6R3B36Q10GZZZ30104	OK	OK	OK	OK	OK	PASS	
		D230520006006		6R3B36Q3UEZZZ30104	OK	OK	OK	OK	OK	PASS	
Hot start	60°C 5	No	Chip_Id	Shipping_No	T0*	T1+F5*				Judgement	Judgement
		D230520006007		6R3B36Q2HCZZZ30104	OK	OK				PASS	PASS
		D230520006008		6R3B36Q3URZZZ30104	OK	OK				PASS	
		D230520006009		6R3B36Q2CJZZZ30104	OK	OK				PASS	
HTS	70°C 500h	No	Chip_Id	Shipping_No	T0*	T100*	T200*	T300*	T500	Judgement	Judgement
		D230520006007		6R3B36Q2HCZZZ30104	OK	OK	OK	OK	OK	PASS	PASS
		D230520006008		6R3B36Q3URZZZ30104	OK	OK	OK	OK	OK	PASS	
		D230520006009		6R3B36Q2CJZZZ30104	OK	OK	OK	OK	OK	PASS	
Cold start	-10°C 5	No	Chip_Id	Shipping_No	T0*	T1+F5*				Judgement	Judgement
		D230520006010		6R3B36Q3UJZZZ30104	OK	OK				PASS	PASS
		D230520006011		6R3B36Q3UBZZZ30104	OK	OK				PASS	
		D230520006012		6R3B36Q2CJZZZ30104	OK	OK				PASS	
LTS	-20°C 500h	No	Chip_Id	Shipping_No	T0*	T100*	T200*	T300*	T500	Judgement	Judgement
		D230520006010		6R3B36Q2CJZZZ30104	OK	OK	OK	OK	OK	PASS	PASS
		D230520006011		6R3B36Q3UJZZZ30104	OK	OK	OK	OK	OK	PASS	
		D230520006012		6R3B36Q2CJZZZ30104	OK	OK	OK	OK	OK	PASS	
140	50%~80% 500h	No	Chip_Id	Shipping_No	T0*	T100*	T200*	T300*	T500	Judgement	Judgement
		D230520006013		6R3B36Q3UJZZZ30104	OK	OK	OK	OK	OK	PASS	PASS
		D230520006014		6R3B36Q2FNZZZ30104	OK	OK	OK	OK	OK	PASS	
		D230520006015		6R3B36Q3URZZZ30104	OK	OK	OK	OK	OK	PASS	
ON/OFF	25°C 30000	No	Chip_Id	Shipping_No	T0*	T180*				Judgement	Judgement
		D230520006016		6R3B36Q2FFZZZ30104	OK	OK				PASS	PASS
		D230520006017		6R3B36Q2HBZZZ30104	OK	OK				PASS	
		D230520006018		6R3B36Q2FLZZZ30104	OK	OK				PASS	
TST	20°C(30min) ~ 60°C(30min), 100cycles judge, 100cycles	No	Chip_Id	Shipping_No	C0*	C100*				Judgement	Judgement
		D230520006016		6R3B36Q2FFZZZ30104	-	-				PASS	PASS
		D230520006017		6R3B36Q2HBZZZ30104	-	-				PASS	
		D230520006018		6R3B36Q2FLZZZ30104	-	-				PASS	
Remark		* # Judge Point / # Judge Point 5) Defect Alarm 5) 6~15) Defect Description									

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EMI Test



	Freq	Level	Read Level	Ant Factor	Limit Line	Over Limit	Remark
	MHz	dBuV/m	dBuV	dB/m	dBuV/m	dB	
1	322.94000	23.57	17.13	6.44	37.00	-13.43	
2	809.88000	26.85	11.62	15.23	37.00	-10.15	
3	PP968.96000	28.54	10.97	17.57	37.00	-8.46	



	Freq	Level	Read Level	Ant Factor	Limit Line	Over Limit	Remark
	MHz	dBuV/m	dBuV	dB/m	dBuV/m	dB	
1	321.97000	23.48	16.62	6.86	37.00	-13.52	
2	809.88000	26.65	10.81	15.84	37.00	-10.35	
3	PP968.96000	32.41	15.21	17.20	37.00	-4.59	
4	QP971.89300	28.32	11.12	17.20	37.00	-8.68	QP

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ESD Test Report Form

☐ Record ☒ ESD check ☐ Customer check

File Name	20230613-5D2360032-G156HAN05.0			Test Ambient: 26 °C / 55% RH			
Application Sec.	RRCAC3	User	黃泓儒	Ext.	86-503004	Sample Received	2023/7/5
Model Name	G156HAN05.0	Quantity (pcs)	3	Panel S/N	6R3B35Q2HSZZ-ZS0104	#1	
					6R3B35Q1DPZZ-ZS0104	#2	
					6R3B35Q2FCZZ-ZS0104	#3	
Examiner	王溪源	Ext.	504202	Date Tested	2023/7/6	Date Completed	2023/7/6
Test standard	☒ AUO Fixed Point Position Rules ☐ AUO IC Nearly Point Position Rules ☐ Customer Rules						
Stage	DVT		Resolution	Frame rate (Hz)	Color depth (bit)	Pixel clock (MHz)	Level B
			1920 X 1080				
XPCB Part №		Layout №		PCB Vendor	N/A		
YPCB Part №		Layout №		PCBA Vendor	N/A		
C/B Part №		Layout №		ASIC			
Inverter Part №	NA	Inverter Vendor	NA	SOURCE IC			
				GATE IC			
Purposes	其他, DVT驗證						
Test Specification: Test Level Requirement: B				Results PASS			
Test Standard: AUO Fixed Point Position Rules (8 points, 25 times/point)							

AUO

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